

● EASY

● ALGEBRA

Linear functions

03

10 questions · ~15 min

Q1

ALGEBRA

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 7 and passes through the point $(0, 5)$. Which equation defines f ?

- A. $f(x) = 5x$
- B. $f(x) = 35x$
- C. $f(x) = 7x + 5$
- D. $f(x) = 12x + 5$

Marisol drove 3 hours from City A to City B. The equation below estimates the distance d , in miles, Marisol traveled after driving for t hours.

$$d = 45t$$

Which of the following does 45 represent in the equation?

- A. Marisol took 45 trips from City A to City B.
- B. The distance between City A and City B is 45 miles.
- C. Marisol drove at an average speed of about 45 miles per hour.
- D. It took Marisol 45 hours to drive from City A to City B.

The function f is defined by $fx = 92x + 3$. For what value of x does $fx = 63$?

- A. 2
- B. 5
- C. 7
- D. 30

The function f is defined by $f(x) = 4x - 3$. What is the value of $f(10)$?

- A. -30
- B. 37
- C. 40
- D. 43

$$d = 16 - \frac{x}{30}$$

The equation shown gives the estimated amount of diesel d , in gallons, that remains in the gas tank of a truck after being driven x miles, where $0 \leq x \leq 480$. What is the estimated amount of diesel, in gallons, that remains in the gas tank of the truck when $x = 300$?

- A. 0
- B. 6
- C. 14
- D. 16

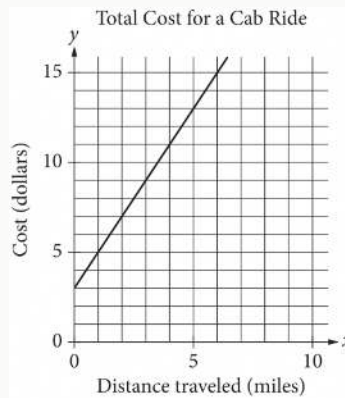
The function f is defined by $f(x) = 25x + 30$. What is the value of fx when $x = 2$?

- A. 50
- B. 57
- C. 80
- D. 110

Josie purchased a pass that cost \$ 19.50 to tour a nature preserve. She also purchased 10 identical stickers to give to her nieces and nephews. The pass and the stickers cost \$ 62.00 total. What was the cost of one sticker?

- A. \$ 1.95
- B. \$ 4.25
- C. \$ 6.20
- D. \$ 8.15

The line graphed in the xy -plane below models the total cost, in dollars, for a cab ride, y , in a certain city during nonpeak hours based on the number of miles traveled, x .



According to the graph, what is the cost for each additional mile traveled, in dollars, of a cab ride?

- A. \$2.00
- B. \$2.60
- C. \$3.00
- D. \$5.00

Q9

GRID-IN ALGEBRA

The function f is defined by the equation $fx = 7x + 2$. What is the value of fx when $x = 4$?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The function f is defined by $fx = 80 - 6x$. What is the value of $f7$?

A. 13

B. 38

C. 74

D. 81